

Strategy for Fabricating Degradable Low-Surface-Energy Cross-Linked Networks with Excellent Anti-Fouling Properties

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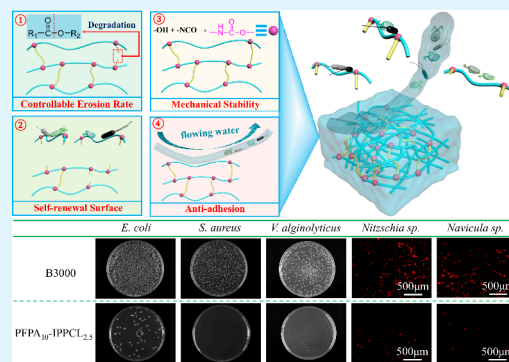
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ABSTRACT: Marine biofouling negatively impacts marine industries and ship navigation. However, current coatings are based on a single antifouling mechanism, which is insufficient to cope with the complex and ever-changing marine environment. Herein, multifunctional antifouling coatings were developed using a material system containing perfluoropolyether and caprolactone chains. First, an acrylic resin containing perfluoropolyether side chains was synthesized as a liquid-repellent component and then a degradable cross-linked network was constructed by bridging polycaprolactone chains. Surprisingly, polycaprolactone chains not only effectively improved the tensile strength but also provided flexibility to the resin. Thus, the coating exhibited satisfactory mechanical stability and low roughness (4.06 nm) during dynamic polishing. It is worth noting that the cross-linked network with a low surface energy (SE) ($22.0 \text{ mJ}\cdot\text{m}^{-2}$) effectively inhibited the adhesion of marine fouling organisms. Moreover, the hydrolysis of ester groups promoted the formation of a self-renewing surface, and the synergistic effect of the low SE and degradability of the coating ensured excellent and long-lasting antifouling performance of the coating. The coating reduced the adhesions of *Vibrio alginolyticus*, *Nitzschia* sp., and *Navicula* sp. by 99.99, 84.6, and 91.0%, respectively, compared with their adhesions to a commercially available self-polishing coating (B3000). Thus, the degradable low-SE antifouling coating produced using the proposed strategy can be potentially applied to various maritime industries.

KEYWORDS: perfluoropolyether, low surface energy, dynamic hydrolysis degradation, mechanical stability, antifouling



1. INTRODUCTION

The importance of the marine transportation industry has become increasingly prominent with the rapid development of global trade. However, marine biofouling has become a key factor restricting the development of the marine industry.^{1–6} Building an antifouling coating on the ship surfaces is considered the most direct and effective method to solve the problem of biofouling. Self-polishing coatings composed of acrylic resin and biocides are considered to be highly effective antibiofouling coatings.⁷ Reportedly, self-polishing coatings based on silyl, copper or zinc acrylates can continuously release biocides via hydrolysis and polishing in seawater.^{8,9} However, the antifouling performance of self-polishing coatings is directly related to the speed of ships.¹⁰ Under static or low-speed navigation conditions, owing to the nondegradability of the resin backbone, the scouring of seawater is insufficient to strip the outermost resin, resulting in inability to effectively release the biocides inside the coating.¹¹ Hence, the static antifouling performance becomes unsatisfactory. Researchers have synthesized various cyclic ketene acetals to address this issue.^{12–14} For example, Jackson et al. synthesized 2-methylene-1,3-dioxepane and copolymerized it with *tert*-butyl acrylate to obtain degradable poly(acrylic acid). The coating degraded by 27.5% within 28 days under environmental

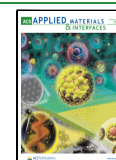
conditions.¹² Galanopoulou et al. added 5,6-benzo-2-methylene-1,3-dioxepane (BMDO) as a comonomer during the polymerization of methyl methacrylate (MMA), thereby introducing degradable ester groups into the poly(methyl methacrylate) backbone and regulating the molecular weight of the degradation products by optimizing the BMDO content.¹³ In addition, using biodegradable materials as self-polishing resins is also a new approach to solving the problem of unsatisfactory static antifouling performance.^{15–17} Ai et al. prepared a degradable network based on poly(ϵ -caprolactone) (PCL), which can decompose into small molecular fragments in seawater, and achieve self-renewal surface without scouring of seawater.¹⁵ Notably, although self-polishing resins offer considerable technical advantages, they do not have antifouling properties. In practical applications, toxic heavy metal biocides are still required to achieve optimal antifouling performance.¹⁸

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To decrease the impact of heavy metal biocides on the marine environment, people are committed to developing various antifouling coating technologies that do not contain heavy metal biocides, including polydimethylsiloxane (PDMS),^{19,20} zwitterionic polymers,^{21,22} amphiphilic polymers,^{23,24} slippery liquid-infused porous surface^{25,26} and hydrogels.^{27,28} Among them, PDMS coatings have garnered widespread attention as they enable the application of low surface energy (SE) coatings in marine antifouling systems because of their low SE and weak adhesion to hard-fouling organisms such as barnacles and mussels.²⁹ However, PDMS coatings also suffer obvious shortcomings, as their low resistance to bacteria and other microorganisms render them ineffective in coping with complex marine environments.³⁰ Therefore, researchers are gradually diverting their attention to other types of low-SE coatings.^{31–33} Li et al. achieved hydrophobic modification of coatings via grafting 1H,1H,2H,2H-perfluoro-1-dodecanol onto polystyrene-*b*-poly(acrylic acid) copolymers and inhibited *Escherichia coli* with >98% reduction in bacterial counts.³¹ Li et al. achieved hydrophobic modification of the resin by mixing perfluoropolyether lubricant with acrylic boron fluorinated polymer via ultrasonic dispersion, and they demonstrated that the lubricating layer formed by perfluoropolyether effectively hindered the adhesion of diatoms and released the adhered diatoms.³² These low-SE coatings exhibited remarkable antifouling and biological adhesion properties. However, low-SE coatings readily lose their antifouling ability once their surface properties are altered by external factors during application.³⁴ This occurs because low-SE coatings, unlike self-polishing coatings, cannot be hydrolyzed and polished in seawater to form self-renewing surfaces. Therefore, developing a coating that combines satisfactory self-polishing performance and low SE characteristics is a challenging task.

In summary, to solve the above-mentioned problems of self-polishing coatings and low-surface-energy antifouling coatings mentioned above, this study designed a self-polishing, antifouling coating with low-SE and without biocides. Perfluoropolyether (PFPE) can effectively inhibit adhesion of fouling organisms considering its low-SE characteristics.^{35,36} Other researchers mainly focus on perfluoropolyether elastomers or incorporating low molecular weight perfluoropolyether oil into resins by means of physical blending. There is relatively little research on use of perfluoropolyether chains to modify acrylic resins and evaluation of their antifouling performance. Therefore, this study first synthesized a series of perfluoropolyether-modified acrylic (PFPA) resins to reduce the SE of the coating. The self-polishing property was achieved by introducing isocyanate-terminated polycaprolactone (IPPCL) into the resin system. IPPCL is degradable in the seawater environment, because of which the coating surface can be continuously polished and renewed during dynamic motion while ensuring its durability and longevity. Noteworthy, the three-dimensional cross-linked network formed via reaction of PFPA and IPPCL effectively increases the mechanical stability of the coating. In addition, by controlling the cross-linking degree of the coating, its erosion rate can be effectively regulated to adapt to various ship navigation requirements. The coating synthesized via the above-mentioned ideas exhibited remarkable antifouling effects on bacteria and algae, particularly with an antiadhesion rate of >99.9% for *Vibrio alginolyticus*.

2. EXPERIMENTAL SECTION

2.1. Materials and Chemicals. Methyl methacrylate (MMA), *n*-butyl acrylate (BA), 2-hydroxyethyl methacrylate (HEMA), 1-dodecanethiol, isophorone diisocyanate (IPDI), ditin butyl dilaurate (DBTDL), 2,2'-azobis(2-methylpropionitrile) (AIBN), xylene (XY) and dichloromethane (DCM) were purchased from Inncochem. Trimethylsilyl acrylate (TMLA) was obtained from Adamas. Polycaprolactone (PCL, Mw 500) was purchased from Jining Huakai Resin Co., Ltd. Perfluoropolyether methacrylate (PFPE–MMA, Mw 600) was obtained from Suzhou Cangmu New Materials Co., Ltd. For comparison, a self-polishing coating (B3000) was procured from Wuhan HESITE Coating Materials Co., Ltd. All chemicals were used as received without further purification.

2.2. Synthesis of IPPCL. Briefly, IPDI (22.23 g), DCM (36.5 g), and DBTDL (0.1%) were heated to 40 °C in a four-neck flask and stirred for 0.5 h. Subsequently, PCL at a specified stoichiometric ratio of $n_{\text{OH}}:n_{\text{NCO}} \approx 1:2$ was introduced dropwise into the flask over a period of 2 h. The product was yielded after 10-h reaction at 40 °C. After removing the solvent via rotary evaporation, isocyanate-terminated polycaprolactone (IPPCL) was obtained.

2.3. Synthesis of PFPA. Initially, the XY solvent was heated to 90 °C in a four-necked flask under a nitrogen atmosphere. Subsequently, a mixture of acrylic monomer, 1-dodecanethiol and AIBN was added dropwise to XY over a period of 2 h. Here, 1-dodecanethiol and AIBN together constituted 1% of the overall monomer amount. After 2 h of heat preservation, a small amount of AIBN was added dropwise within 1 h, and the reaction continued for an additional 3 h. The crude product was precipitated in *n*-hexane, then redissolved in XY and reprecipitated in *n*-hexane. This procedure was performed three times to obtain PFPA. PFPA samples with different ratios of ingredients were prepared, as shown in Table 1. Each sample is subscripted with its PFPE–MMA mass fraction (for example, PFPA₅ denotes a PFPE–MMA mass fraction of 5% in PFPA).

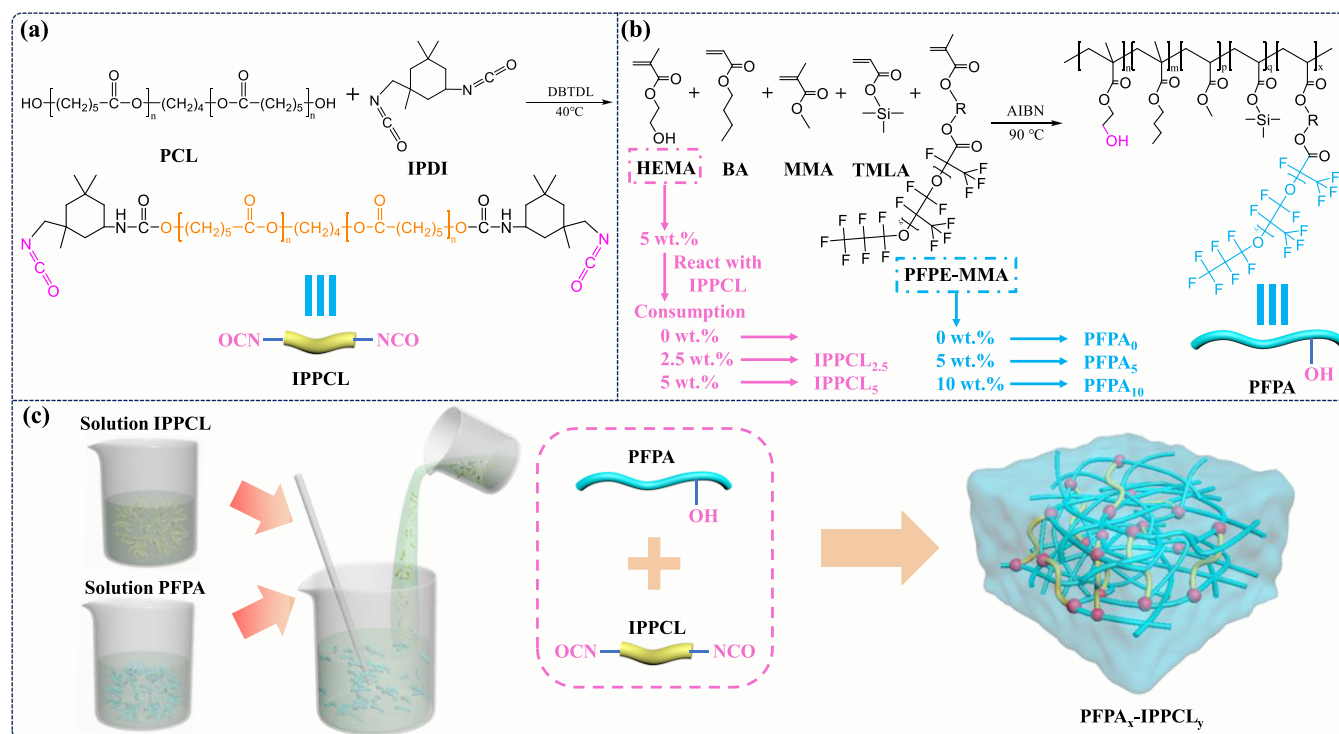
Table 1. Ingredient Ratios in the PFPA Sample

	MMA	BA	HEMA	PFPE–MMA	TMLA
PFPA ₀	30%	35%	5%	0%	30%
PFPA ₅	25%	35%	5%	5%	30%
PFPA ₁₀	20%	35%	5%	10%	30%

2.4. Preparation of Degradable Polyurethane Coatings (PFPA_x–IPPCL_y). First, PFPA and IPPCL were dissolved in XY and DCM (40 wt %), respectively. The PFPA and IPPCL solutions were mixed at their specified stoichiometric ratios ($n_{\text{OH}}:n_{\text{NCO}} = 1:1–1.05$, $1:0.5–0.525$ and $1:0$). Finally, coatings with a dry thickness of $\sim 250 \mu\text{m}$ were prepared via solution casting. The coatings were cured at 40 °C for 3 days, dried at 70 °C for 10 h to completely remove the solvent and then applied to 316L stainless steel for water absorption tests, coating erosion tests, surface topography observations, X-ray photoelectron spectroscopy (XPS) and antifouling performance tests. Further, the coatings were applied to the epoxy fibreboard surface for contact-angle measurements and adhesion tests. The epoxy fibreboard can be used directly for measurements, but 316L stainless steel must be degreased and acid picked prior to use.

As shown in Table 1, the mass percentage of HEMA in all PFPA_x samples is 5%. As IPPCL primarily reacts with HEMA in PFPA, the degradable low-SE cross-linked network coatings are named PFPA_x–IPPCL_y, where *y* denotes the proportion of HEMA participating in the cross-linking reaction in PFPA. When PFPA and IPPCL are mixed at the specified stoichiometric ratio of $n_{\text{OH}}:n_{\text{NCO}} = 1:1–1.05$, 5 wt % of HEMA participates in the reaction; hence, the coating is named PFPA_x–IPPCL₅. When PFPA and IPPCL are mixed at the specified stoichiometric ratio of $n_{\text{OH}}:n_{\text{NCO}} = 1:0.5–0.525$, 2.5 wt % of HEMA participates in the reaction; therefore, the coating is named PFPA_x–IPPCL_{2.5}. For comparison, a PFPA₁₀ coating with no IPPCL was also prepared. In summary, the subscripts *x* and *y* in the name 'PFPA_x–IPPCL_y' denote the mass fraction of PFPE–MMA in PFPA and of

Scheme 1. Synthesis Process of the (a) Isocyanate-Terminated Polycaprolactone and (b) Perfluoropolyether-Based Acrylic Resin. (c) Schematic Showing the Fabrication Process of the Degradable Low-Surface-Energy Coating



HEMA participating in the cross-linking reaction of PFPA, respectively.

2.5. Characterization of Composition and Structure. The ¹H and ¹⁹F nuclear magnetic resonance (NMR) spectra were collected on a Bruker Avance III HD spectrometer (500 MHz) with DMSO-*d*₆ as the solvent. Fourier-transform infrared (FT-IR) spectra were obtained in the wavenumber range of 4000–400 cm^{−1} on a Nicolet 6700 FT-IR spectrometer (Thermo Fisher, USA) equipped with a diamond attenuated total reflectance window. The hydroxyl content of the synthesized PFPA was determined following GB/T 12008 guidelines. XPS was performed on a Thermo escalab 250XI XPS spectrometer (Thermo Electron, USA). For this, samples were irradiated with monochromatic Al Kα X-rays with a spot diameter of 650 mm and a power of 14.8 kV. The full-spectrum and narrow-spectrum energy ranges were set to 100 and 20 eV, with step sizes of 1.0 and 0.1 eV, respectively. All binding energies were calibrated using the C 1 s peak at 284.8 eV. The molecular weights and distributions of PFPA were determined using Gel Permeation Chromatography (GPC) with a Shimadzu LC-20AD system equipped with PLgel columns (Shodex KF804L, Japan) and a differential refractometer detector (RID-20, Japan). The analysis was conducted in tetrahydrofuran (THF) at 35 °C with an eluent flow rate of 1.0 mL/min. Calibration of the GPC system was performed using polystyrene standards ranging from 50 to 1 × 10⁶ g/mol with narrow polydispersity indices.

2.6. Mechanical and Thermodynamic Test. Samples for tensile tests were prepared according to the GB/T 2567-2021 standard. The tests were performed at least three times, and the results were reported as averages with standard deviations. The tensile properties of coatings were assessed using a universal mechanical tester (Instron 8871 W, China) at room temperature. The adhesion strengths of coatings applied on the epoxy fibreboard surface were measured with a fully automatic pull-out adhesion tester (BGD 500/S, China) according to ASTM D4541. Each sample was measured thrice, and the results were presented as averages with standard deviations.

Thermogravimetric analysis (TGA) was performed using a simultaneous thermal analyzer (STA 449 F3 Jupiter, Germany) under a nitrogen atmosphere. Samples were heated from room

temperature to 600 °C at a heating rate of 10 °C·min^{−1}. Additionally, the differential thermogravimetric (DTG) curves were obtained.

2.7. Contact Angle and Surface Free Energy Measurement.

The static water contact angle was determined with a drop shaped analyzer DSA 25 instrument (Krüss, Germany) using a 5 μL DI water droplet at room temperature (25 °C). Each sample was measured five times to obtain the average value with standard deviation. The static kerosene and hexadecane contact angles of the coating surface were measured using the same protocol.

The surface free energies of coatings were calculated using the Owens–Wendt method [eqs 1 and 2].

$$\gamma_{LV} + \cos \theta = 2(\gamma_{SV}^d \gamma_{LV}^d)^{1/2} + 2(\gamma_{SV}^p \gamma_{LV}^p)^{1/2} \quad (1)$$

$$\gamma_{SV} = \gamma_{SV}^p + \gamma_{SV}^d \quad (2)$$

where γ_{LV} and γ_{SV} represent interfacial tensions at the liquid–vapor and solid–vapor interfaces, respectively; superscripts *d* and *p* respectively denote the dispersion and polar components of the surface free energy and θ is the static contact angle at the solid–liquid–vapor interface of deionized (DI) water and hexadecane, whose surface tensions are known.

2.8. Atomic Force Microscopy Imaging and Force Measurements. The surface topographies and root mean squares of the surface roughness heights (*R_q*) of coatings were investigated using atomic force microscopy (AFM) (Bruker Dimension Icon system, Rheinstetten, Germany). AFM was operated in the standard tapping mode in air, and the images were recorded with scan sizes ranging from 20 × 20 μm at scanning rates of 1 Hz and 512 samples/line. In addition, silicon nitride AFM probes with a nominal spring constant of 0.06 N/m and a tip with a radius of curvature of 20 nm was used for AFM force measurements.

2.9. Coating Erosion and Water Absorption Experiments.

Coating samples were immersed in DI water for 72 h to obtain their water absorption data, which were calculated according to eq 3.

$$\text{water absorption(\%)} = (M_{\text{wet}} - M_{\text{dry}}) / (M_{\text{dry}} - M_{\text{substrate}}) \times 100\% \quad (3)$$

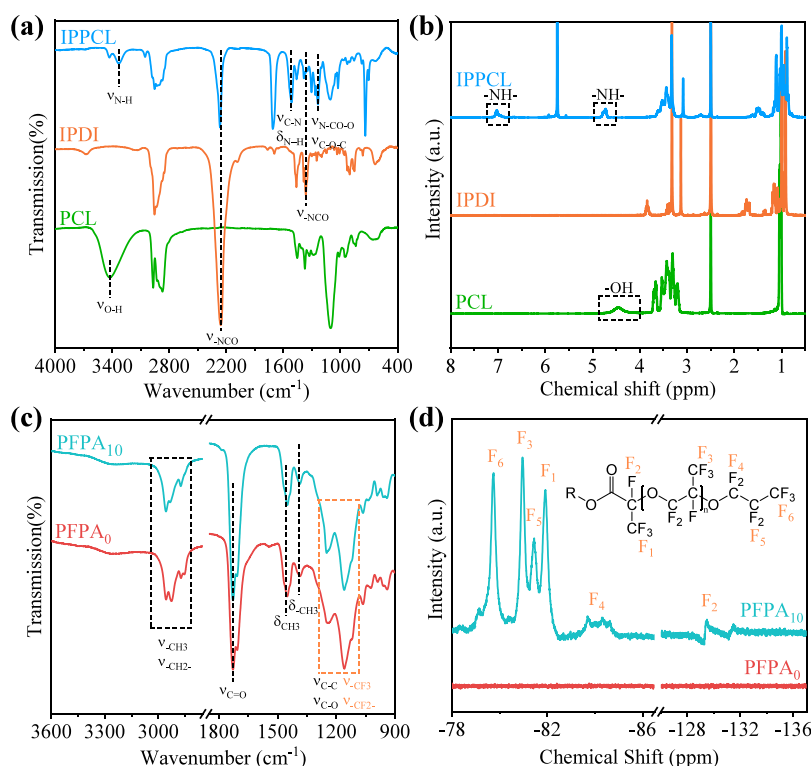


Figure 1. (a) FT-IR and (b) ^1H NMR spectra of IPPCL, IPDI, and PCL; (c) FT-IR and (d) ^{19}F NMR spectra of PFPA.

where M_{wet} , M_{dry} , and $M_{\text{substrate}}$ denote the masses of the coated sample filled with water, coated sample after drying and substrate of the coated sample, respectively.

The coating erosion experiments were performed on the device shown in Figure S1. Coated samples were immersed in 4 L of simulated seawater, placed 20 cm from the center of a stirring paddle with a blade size of 10 cm. Subsequently, the erosion data were recorded at 5-day intervals under a stirring speed of 1000 r/min. The masses of coated samples were also recorded every 5 days. To determine the mass loss, samples were completely dried in the oven at 70 °C for 10 h before weighing. The mass loss was then calculated as eq 4.

$$\text{mass loss} = (M_0 - M_n)/S \quad (4)$$

where M_0 represents the mass of the coated sample before coating erosion test, M_n represents the mass of the coated sample after n days of erosion ($n = 5, 10, 15, 20$ and 25) and S is the area of the coated sample.

2.10. Antifouling Performance Evaluation. *V. alginolyticus* was selected as representative marine bacterium, and the antibacterial-adhesion performance of coatings was evaluated using the plate counting method. In this test, 100 L of bacterial suspension (10^8 CFU/mL) were dropped onto the surface of a sterilized coated sample and a sterilized cover film ($1.6 \text{ cm} \times 1.6 \text{ cm}$) was applied to the coating surface, ensuring that the sample and bacterial solution were evenly contacted within the edges of the cover film. The sample was then incubated in a constant-temperature incubator at 37 °C for 24 h. After cultivation, the sample was gently rinsed with sterile PBS solution three times to remove any unattached bacteria and then transferred to a centrifuge tube. After adding 10 mL of sterile PBS solution, the tube was sonicated for 8 min and then vortexed for 15 s. The sonicated bacterial solution was diluted 10 times continuously with sterile PBS solution. Next, 100 μL of each diluted solution was evenly applied to the LB solid culture medium. Further, the sample was incubated at 37 °C for 18 h, and the number of adherent colonies was recorded.

Nitzschia sp., *Navicula* sp. and *Chlorella* sp. were selected as representative algae, and the antifouling adhesion performance of

coatings was observed and evaluated using an inverted fluorescence microscope (IX73-DP80, Japan). In this method, coated samples were immersed in a six-well plate containing 15 mL of the algal suspension and incubated in an artificial climate chamber at a light to dark ratio of 12 to 12 h for 24 h. The surfaces of coated samples were gently rinsed with DI water to remove unattached algae. The ImageJ software was used to determine the algal coverage on the sample surface, and each sample was measured thrice to obtain the average value with standard deviation. Furthermore, the commercially available B3000 self-polishing coating was employed as blank control for the antibacterial-adhesion and antifouling-adhesion tests to provide a visual comparison of the antifouling performance of the prepared coatings.

3. RESULTS AND DISCUSSION

3.1. Characterization of PFPA, IPPCL, and PFPA_x-IPPCL. Scheme 1 illustrates the synthesis route for IPPCL, PFPA and PFPA_x-IPPCL. First, IPPCL was effectively synthesized through the reaction between the alcohol hydroxyl group of PCL and isocyanate group of IPDI (Scheme 1a). Figure 1a illustrates the FT-IR spectra of PCL, IPDI and IPPCL. The absorption peak at 3331 cm^{-1} is attributable to N-H stretching vibrations,³⁷ and the peak at 1515 cm^{-1} corresponds to C-N stretching and N-H bending vibrations.³⁸ Notably, the hydroxyl stretching vibration peak of PCL at 3426 cm^{-1} is not visible in the spectrum of IPPCL, indicating that the active hydroxyl group of PCL reacted with the isocyanate group of IPDI to form a urethane bond. Furthermore, at 2267 cm^{-1} , the stretching vibration peak of -NCO in the IPPCL spectrum clearly appeared,³⁹ confirming the presence of isocyanate groups that can participate in the subsequent cross-linking reaction. The structure of IPPCL was further clarified through ^1H NMR measurements (Figure 1b). The peak at 4.5 ppm disappeared after heavy water exchange in the spectrum of PCL (Figure S2), indicating that the proton was an active hydrogen belonging to -OH in PCL. The

IPPCL spectrum also showed no characteristic peak at 4.5 ppm but presented new peaks at 7.02 and 4.72 ppm, which were assigned to protons of the urethane group. The active hydrogen of the urethane group disappeared after heavy water exchange (see the ^1H NMR spectrum in Figure S3). Moreover, as shown in Table 2, the molecular weight and distribution of IPPCL are higher than those of PCL. These findings clarify the successful synthesis of IPPCL.

Table 2. Molecular Weights and Distributions of PFPA, PCL, and IPPCL

sample	M_w (g·mol $^{-1}$)	PDI
PCL	490	1.05374
IPPCL	1149	1.15161
PFPA $_0$	4959	2.14318
PFPA $_5$	4582	1.96932
PFPA $_{10}$	4388	1.83503

Next, a series of PFPA resins were synthesized via the free-radical polymerization of MMA, BA, HEMA, PFPE–MMA and TMLA in xylene (Scheme 1b). Among them, MMA, as a hard monomer, is mainly used to improve the hardness of coatings, while BA is a soft monomer, primarily utilized to improve flexibility. Meanwhile, hydrolyzable silicone ester bonds in TMLA can generate carboxyl groups after hydrolysis, accelerating the hydrolysis of ester groups in the IPPCL chain. HEMA, which contains hydroxyl groups, can participate in subsequent reactions with IPPCL. Finally, PFPE–MMA introduces hydrophobic PFPE chains into the coating. The GPC curve of PFPA resins are displayed in Figure S4 and molecular weights and distributions of PFPA resins are shown in Table 2. The hydroxyl content of the synthesized PFPA was found to be approximately 0.64%, which was consistent with the experiment value. Figure 1c displays the FT-IR spectra of PFPA. The peaks at 2957, 2874, 2933, and 2857 cm^{-1} correspond to the characteristic asymmetrical and symmetric stretching vibration peaks of $-\text{CH}_3$ and $-\text{CH}_2-$ groups, respectively. The characteristic peaks at 1449 and 1386 cm^{-1}

correspond to the asymmetric and symmetric angular vibrations of $-\text{CH}_3$ groups, respectively.⁴⁰ The peak at 1727 cm^{-1} is ascribed to the stretching vibrations of $\text{C}=\text{O}$ groups.⁴¹ The broad absorption peak at 3262 cm^{-1} is attributable to the $-\text{OH}$ group,⁴² which can participate in subsequent cross-linking reactions. The absorption peaks at 1249 and 1159 cm^{-1} belong to the asymmetric stretching vibrations of the $\text{C}-\text{C}$ and $\text{C}-\text{O}$ groups, respectively. Further, the absorption peaks at 1249 and 1159 cm^{-1} can also belong to the symmetric and antisymmetric stretching vibrations of the $-\text{CF}_2-$ and $-\text{CF}_3$ groups, respectively.⁴³ Thus, the absorption peaks at 1249 and 1159 cm^{-1} cannot demonstrate the copolymerization of PFPE–MMA with other acrylic monomers because the $-\text{CF}_2-$ and $-\text{CF}_3$ groups are much less abundant than the $\text{C}-\text{C}$ and $\text{C}-\text{O}$ groups. Consequently, copolymerization is confirmed in the ^{19}F NMR spectra of PFPA $_0$ and PFPA $_{10}$ (Figure 1d). The characteristic peaks of F are notably absent over the tested wavelength range of PFPA $_0$; however, the spectrum of PFPA $_{10}$ exhibits multiple peaks attributable to the diverse chemical environments of different F atoms. Thus, all peaks in the PFPA $_{10}$ spectrum are successfully assigned,⁴⁴ demonstrating that the PFPE–MMA monomer can copolymerise with other acrylic monomers to form long PFPA chains.

Finally, IPPCL and PFPA were uniformly mixed to prepare low-SE antifouling coatings (Scheme 1c). Figure S5 displays the FT-IR spectrum of the PFPA $_x$ –IPPCL $_y$. The characteristic peaks in the PFPA $_x$ –IPPCL $_y$ spectrum are similar to those in the PFPA spectrum because PFPA $_{10}$ is the primary component of PFPA $_x$ –IPPCL $_y$. Notably, the characteristic absorption peak of the $-\text{NCO}$ group at 2267 cm^{-1} is absent, indicating that the isocyanate group of IPPCL successfully reacted with the hydroxyl group of PFPA. The novel absorption peak at 1533 cm^{-1} corresponds to the stretching vibrations of $\text{C}-\text{N}$, confirming the formation of urethane groups. The above results demonstrate the successful preparation of PFPA $_x$ –IPPCL $_y$.

3.2. Erosion Performance of Coatings. As a result of the progressive degradation and cleavage of polycaprolactone polymer chains upon exposure to seawater, the thickness of

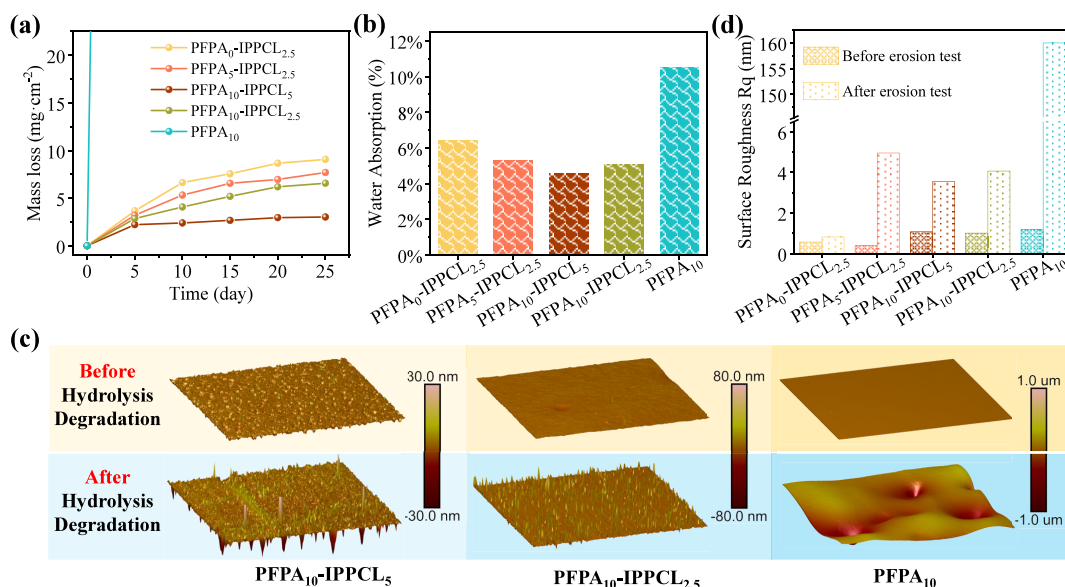


Figure 2. (a) Erosion rate curves and (b) water absorptions of the coatings; (c) surface topographies and (d) surface roughness values of the coatings before and after the erosion test.

the antifouling coating decreases as the ship navigates. To quantify this phenomenon, the erosion rates of coatings were characterized in simulated seawater using an erosion coating test device. PFPA₁₀ exhibited an unusually rapid erosion rate, losing 61.3 mg·cm⁻² of its mass on the first day (Figure 2a). Such a high erosion rate reduces the long-term stability and effectiveness of antifouling coatings. The erosion rate of PFPA₁₀-IPPCL_{2.5} was considerably slower (only 6.56 mg·cm⁻² over the 25-d testing period). The surface chains of uncured PFPA₁₀ were continuously stripped due to the scouring of the simulated seawater; on the contrary, the surface chains of IPPCL-cured PFPA₁₀ were limited by the cross-linked network formed between IPPCL and PFPA₁₀. Under the scouring of simulated seawater, the stripping process must overcome not only the intermolecular forces between the original chains but also the hydrolysis of the ester groups of IPPCL; accordingly, the erosion rate decreases. Figure S6 shows the FT-IR spectrum of PFPA₁₀-IPPCL_{2.5} after erosion test. The broad peak in the 3000–3600 cm⁻¹ range corresponds to the stretching vibrations of the -OH group,⁴⁵ indirectly indicating the hydrolysis degradation of the coating. Furthermore, increasing the IPPCL content increased the cross-linking degree, thereby enhancing the number of IPPCL chains connected to a single PFPA₁₀ chain. Increasing the number of IPPCL chains during hydrolysis prolonged the waiting period for the release of PFPA₁₀ chains from the coating surface, leading to a notable decrease in the erosion rate. Thus, PFPA₁₀-IPPCL₅ displayed the slowest erosion rate, with a dynamic weight loss of 3.01 mg·cm⁻² after 25 d. Enhancing the hydrophobicity of the coating also decreased the erosion rate (which is later discussed in Section 3.5). Following the 25-d testing period, the measured masses losses of PFPA₅-IPPCL_{2.5} and PFPA₀-IPPCL_{2.5} were 7.69 and 9.07 mg·cm⁻², respectively, slightly higher than that of PFPA₁₀-IPPCL_{2.5}. In addition, the erosion rate of the coatings was initially high but tended to stabilize after 10 d.

The internal factors influencing the erosion rates of the coatings were clarified through water absorption tests of the coatings (Figure 2b). The water absorptions of PFPA₀-IPPCL_{2.5}, PFPA₅-IPPCL_{2.5}, PFPA₁₀-IPPCL₅, PFPA₁₀-IPPCL_{2.5} and PFPA₁₀ were 6.41%, 5.30%, 4.57%, 5.07% and 10.52%, respectively. Obviously, the water absorptions of the coatings decreased with increasing PFPE content. In addition, the cross-linked network formed by the interacting IPPCL and PFPA chains effectively prevented the ingress of water molecules and reduced the water storage capacity of the coatings. Consequently, increasing the cross-linking degree reduced the water absorption. The results indicate a positive correlation between the erosion rate and the water absorption of the coating.

A smooth surface is essential for minimizing the drag resistance of a coating during maritime navigation.⁴⁶ The macroscopic images of the coatings before and after erosion test are shown in Figure S7. The PFPA₁₀ surface was noticeably wrinkled; however, the macroscopic morphologies of the other coatings exhibited negligible variations. To deeply understand the subtle alterations on the coating surfaces, surface topography and surface roughness data were obtained via AFM (Figure 2c,d and Figure S8). Prior to the erosion test, all coating surfaces were remarkably smooth with surface roughness (*R_q*) values below 2 nm. The addition of PFPE and IPPCL did not appreciatively alter the initial coating roughness. After erosion test, the roughness values of all

coatings slightly increased due to water absorption, swelling and dehydration shrinkage. More importantly, hydrolysis and polishing induced a dramatic roughness increase on PFPA₁₀ (from 1.18 to 160 nm) and a relatively minimal roughness increase on PFPA₁₀-IPPCL₅ (from 1.08 to 3.55 nm) and PFPA₁₀-IPPCL_{2.5} (from 1.01 to 4.06 nm). This difference can be attributed to IPPCL, which effectively restricts the chain mobility and ensures the mechanical stability of coatings. Consequently, the coating is smooth and can resist the strong shear forces generated during erosion test at high rotational speeds, thus reducing the surface drag.

3.3. Mechanical Properties of Coatings. The surfaces of antifouling coatings must endure considerable loads during high-speed ocean voyages. Therefore, mechanical stability is a prerequisite of antifouling coatings with long-term stability. The impacts of the cross-linking degree and PFPE content on the mechanical properties of PFPA_x-IPPCL_y coatings were investigated via tensile tests. The results are presented in Figure 3a,b. As anticipated, IPPCL successfully linked the

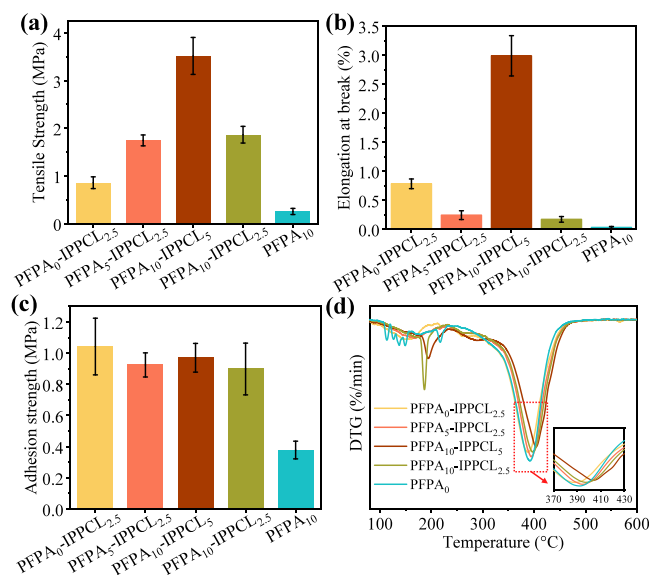


Figure 3. (a) Tensile strength, (b) elongation at break, (c) adhesion strength, and (d) differential thermogravimetric curves of the coatings.

PFPA chains and substantially increased the tensile strength. The toughness of the resulting coating met the requirements of harsh marine environments.^{47,48} IPPCL elevated the tensile strength and elongation at break from 0.255 MPa and 0.029% in PFPA₁₀ to 3.52 MPa and 2.99% in PFPA₁₀-IPPCL₅, respectively. This improvement can be attributed to the flexibility conferred by the low-steric-hindrance -CH₂- group in IPPCL.⁴⁹ Intriguingly, by comparing the data of PFPA_x-IPPCL_{2.5}, increasing the PFPE content in coatings increased the tensile strength but decreased the elongation at break. These results indicate that the mechanical properties of coatings can be effectively optimized through the careful adjustment of the polymer component and cross-linking degree.

Coating-substrate adhesion strength is widely recognized as a critical performance indicator of antifouling coating systems. Consequently, the adhesion strength between the coatings and an epoxy fibreglass panel was examined using an adhesion tester. The adhesion strengths of PFPA₀-IPPCL_{2.5}, PFPA₅-

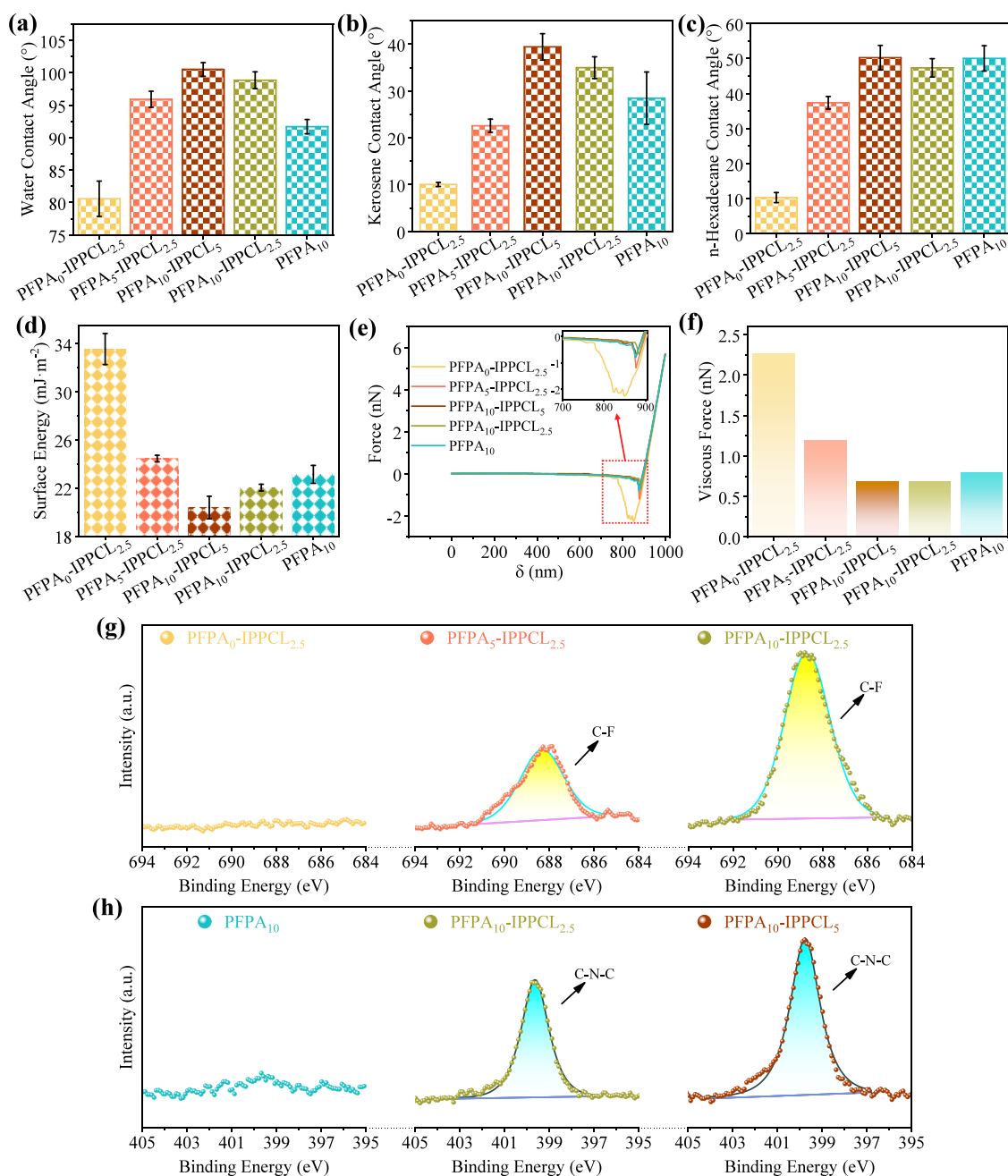


Figure 4. Static contact angles of (a) water, (b) kerosene, and (c) *n*-hexadecane on the coating surface; (d) surface energies, (e) force–displacement curves, and (f) viscous forces of the coating surface; high-resolution (g) F 1s and (h) N 1s XPS spectra of the coatings.

IPPCL_{2.5} and PFPA₁₀-IPPCL_{2.5} were 1.042, 0.924, and 0.898 MPa, respectively (Figure 3c). These results suggest a limited effect of the perfluoropolyether chains on adhesive strength; more specifically, increasing the PFPE content only slightly reduced the interfacial coating–substrate bonding. Much more important is the role of hydrogen bonds in adhesion. The adhesion strength of PFPA₁₀ was only 0.378 MPa, indicating that the urethane groups formed through the reaction between IPPCL and PFPA were rich in hydrogen bonds that enhanced the coating–substrate bonding properties.^{50,51} As the perfluoropolyether component did not notably weaken the adhesion of the coating, IPPCL imparted an excellent adhesive strength that will benefit the long-term application of the coating in complex marine environments.

The contribution of the cross-linked network to the stability of the coating structure was further confirmed based on the thermogravimetric curves of the coatings (Figure S9). The differential thermogravimetric curves and their magnified views in the 370 °C–430 °C range are shown in Figure 3d and its inset, respectively. The maximum thermal decomposition rates of PFPA₀-IPPCL_{2.5}, PFPA₅-IPPCL_{2.5}, PFPA₁₀-IPPCL₅, PFPA₁₀-IPPCL_{2.5} and PFPA₁₀ were observed at 388 °C, 394.7 °C, 404.2 °C, 397.8 and 392.5 °C, respectively. Notably, the maximum thermal decomposition rate temperature increased with increasing the cross-linking degree of coatings, indicating that the cross-linked network formed by IPPCL and PFPA effectively delayed chemical-bond breakage and enhanced the thermal degradation stability of coatings.⁵² Thus, increasing the cross-linking degree increases the energy

(i.e., temperature) at which coatings degrade. Note that PFPE also improved the thermal stability of the coatings, as evidenced by the rise in the temperature of the maximum thermal decomposition rate with increasing PFPE content. This enhancement was attributed to the higher bonding energy of the C–F groups (485 kJ/mol) compared to those of the C–C groups (332 kJ/mol) and C–H groups (414 kJ/mol).⁵³

3.4. Surface Characteristics of Coatings. The surface properties of the coatings play a vital role in their antifouling performance because the coating surface is the adhesion site of marine fouling organisms. The antifouling coating developed in this study predominantly leverages the low-SE properties to deter the adhesion of fouling organisms. Thus, the static water contact angles (WCAs), kerosene contact angles (KCAs) and *n*-hexadecane contact angles (HCAs) were measured to determine the hydrophilic, hydrophobic, oleophilic and oleophobic properties of the PFPA_{*x*}–IPPCL_{*y*} coatings and hence clarify their antifouling mechanism. Figure S10 displays the outlines of the different liquid media on the coating surfaces. Increasing the PEPE content increased the static WCA of PFPA_{*x*}–IPPCL_{2.5} from 80.6° to 98.9°, the static KCA from 10.1° to 35.0° and the static HCA from 10.3° to 47.3° (Figure 4a–c). These findings suggest that increasing the surface content of lyophobic perfluoropolyether chains effectively improves the hydrophobicity and oleophobicity of the coatings. Meanwhile, the cross-linking degree exerted a similar influence on the PFPA content. Increasing the IPPCL content further improved the contact angles of coatings. Meanwhile, increasing the cross-linking degree narrowed the gaps between the polymer chains, effectively inhibiting the penetration of solvent into the coating. Consequently, increasing the IPPCL content enlarged the contact angles of the coatings, implying that hydrophobicity and oleophobicity increased. Figure 4d shows the SEs of PFPA_{*x*}–IPPCL_{*y*} calculated based on the WCAs and HCAs. Opposing the trends of the contact angles, the SEs of the PFPA_{*x*}–IPPCL_{*y*} coatings decreased with increasing PFPE and IPPCL contents. Notably, PFPA₁₀–IPPCL₅ exhibited the lowest SE (~20.4 mJ·m^{−2}, compared to ~33.6 mJ·m^{−2} on PFPA₀–IPPCL_{2.5}). It is concluded that the synergistic effect of PFPA and IPPCL further reduced the SE of the cross-linked network. Meanwhile, the SEs of PFPA₅–IPPCL_{2.5}, PFPA₁₀–IPPCL_{2.5} and PFPA₁₀–IPPCL₀ were 24.5, 22.0, and 23.1 mJ·m^{−2}, respectively. Thus, the excellent antifouling performance of these low-SE coatings can be attributed to their fine liquid repellency. Furthermore, note that the contact-angles and SE variations were consistent with the XPS analysis. Specifically, the augmentation of liquid-repellent components on the coating surface, along with the enhanced cross-linking degree, contributed to liquid contact-angle elevation and SE reduction.

The adsorption and aggregation of fouling cells on the coating surface can be modulated by various intermolecular interactions, including specific interactions such as molecular recognition between ligands and receptors and nonspecific interactions such as hydrogen bonds, hydrophobic forces, van der Waals forces and electrostatic forces.⁵⁴ Consequently, the adhesion of marine fouling organisms can be comprehended by acquiring mechanical information on the coating surfaces. To further elucidate the impact of the lyophobic component content and cross-linking degree on the antifouling-adhesion performance of the coatings, the force–displacement curves of the coating surfaces were obtained via AFM. As depicted in Figure 4e,f, the indentation and withdrawal curves of the probe

on the coating surfaces did not completely overlap, indicating the presence of force hysteresis. The force hysteresis was attributed to the viscoelasticity of the coating, which arose from energy dissipation associated with the relative motion between the probe and the coating during deformation. Based on the lowest point of the force curve, which reflects the interaction force between the coating and the probe, the viscous forces on the PFPA₀–IPPCL_{2.5}, PFPA₅–IPPCL_{2.5}, PFPA₁₀–IPPCL₅, PFPA₁₀–IPPCL_{2.5} and PFPA₁₀–IPPCL₀ surfaces were determined as 2.26, 1.189, 0.6856, 0.6883, and 0.7937 nN, respectively. The viscous forces acting on coatings varied similarly to SEs, indicating that they depend on the PFPE content and cross-linked network degree. Specifically, the viscous force decreased with increasing the PFPE content and was further reduced with increasing the cross-linked network degree. In general, fouling organisms cannot easily adhere to a coating surface with low-viscous forces.⁵⁵

To further elucidate the underlying reasons for the changes in surface characteristics of the coatings, the surface compositions of PFPA_{*x*}–IPPCL_{*y*} were thoroughly investigated. The XPS full scan spectra for PFPA₀–IPPCL_{2.5}, PFPA₅–IPPCL_{2.5}, PFPA₁₀–IPPCL_{2.5}, PFPA₁₀–IPPCL₅, and PFPA₁₀ are shown in Figure S11, while the high-resolution spectra for C 1s, O 1s, F 1s, and N 1s are presented in Figure 4g,h and Figures S12 and S13, respectively. The quantitative analysis results for the atomic fractions in the probed surface regions are summarized in Table S1. Figure S11 indicates that PFPA₅–IPPCL_{2.5}, PFPA₁₀–IPPCL₅, and PFPA₁₀–IPPCL_{2.5} contain C, O, Si, F, and N, while PFPA₀–IPPCL_{2.5} contains only C, O, Si, and N, and PFPA₁₀ contains C, O, Si, and F. The presence of F confirms the enrichment of perfluoropolyether chains on the coating surfaces, while the presence of N indicates the formation of urethane groups. The primary elements in all samples are C and O. The high-resolution XPS spectra for C 1s of the samples are shown in Figure S12. Each C 1s spectrum exhibits four distinct peaks. The peak at 284.8 eV corresponds to the C–C bonds in the resin backbone and side chains. The peak at 285.7 eV is typically attributed to the C–O and C–N bonds in ester and urethane groups, while the peak at 286.7 eV corresponds to the C=O bond in these groups. The peak at approximately 288.9 eV corresponds to O=C–O in both ester and urethane groups. Notably, the introduction of perfluoropolyether chains and the formation of urethane groups slightly negatively shifted the peaks corresponding to C–O and C–N bonds, although the effect on the positions of the C=O and O=C–O peaks is negligible. Furthermore, the formation of urethane groups led to a slight increase in the peak intensity of C–O and C–N bonds (Table S1). In the O 1s spectra shown in Figure S13, the peaks at 532.1 and 533.4 eV correspond to the C–O and C=O bonds in the ester and urethane groups, respectively. Notably, the coatings PFPA₀–IPPCL_{2.5}, PFPA₅–IPPCL_{2.5}, PFPA₁₀–IPPCL_{2.5}, and PFPA₁₀ contain hydroxyl groups that do not participate in the reaction; however, their low concentration resulted in no significant peaks. Furthermore, no obvious regularity can be identified regarding the impact of urethane group formation and the introduction of perfluoropolyether chains on peak position and intensity in the coatings. The peak at ~689 eV was attributed to the C–F groups within the perfluoropolyether chain.⁵⁶ Notably, the intensity of the F 1s peak gradually increased with increasing PFPE content, suggesting that the fluorine-containing chains with low SE tended to migrate toward the coating surface. The enrichment of F content on the surfaces

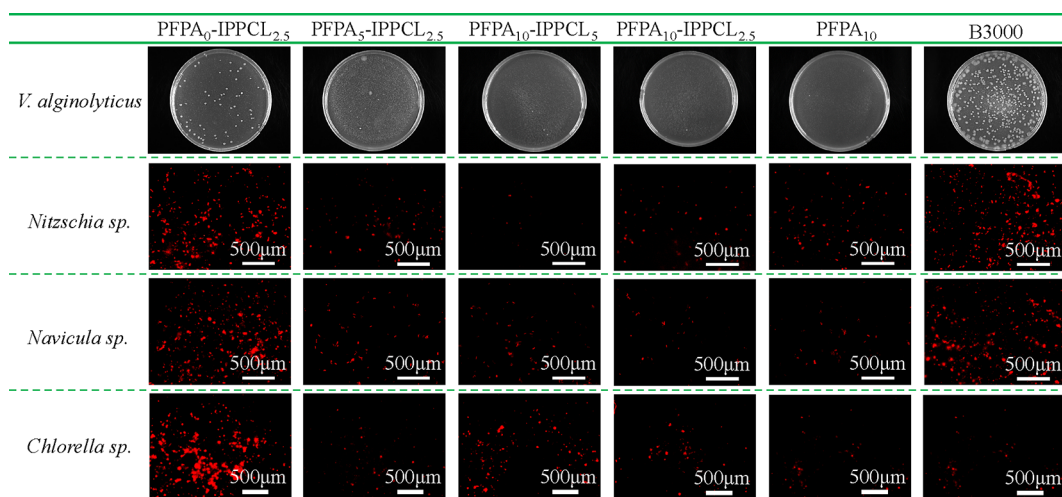


Figure 5. Distributions of *V. alginolyticus* on the LB medium and distributions of *Nitzschia* sp., *Navicula* sp., and *Chlorella* sp. algal cells on coating surfaces.

of the PFPA_x-IPPCL_{2.5} coatings was directly correlated with the PFPE content.^{57,58} As expected, no F 1s peak was detected in the spectrum of the PFPA₀-IPPCL_{2.5} coating as fluorocarbons were absent in this coating. The single peak at ~400 eV was the characteristic peak of C–N groups,⁵⁹ confirming the covalent cross-linking between the hydroxyl and isocyanate groups. Further, the spectrum of PFPA₁₀ exhibited no N 1s peak, and the peak intensity of the C–N groups increased with increasing IPPCL content, indicating an elevated degree of cross-linking.

3.5. Antifouling Performance. Antifouling performance is an important function of marine protective coatings. The initial stage of marine fouling is closely tied to biofilm formation, which is fundamentally associated with the reversible adhesion of bacteria on the coating surface.⁶⁰ Therefore, biofilm formation can be prevented by inhibiting bacterial adhesion on the coating. With their low-SE characteristics, the developed coatings are expected to exhibit high antibacterial-adhesion performance. The antibacterial-adhesion performance of the coatings was evaluated through antibacterial assays based on the plate counting method, where the commercially available self-polishing coating B3000 was used as the control. The representative bacterium, *V. alginolyticus*, which is widely found in the marine environment, was employed in the assays. The number of bacterial colonies on the coating surface provided a visual interpretation of the antibacterial-adhesion performance. The distribution of bacterium coated on the Luria–Bertani (LB) medium after 24 h of incubation with different coatings is shown in Figure 5, and the bacterial densities are shown in Figure 6a. High bacterial colony densities were observed on the control and PFPA₀-IPPCL_{2.5} surfaces, indicating limited antibacterial activity (*V. alginolyticus* densities were $\sim 5.4 \times 10^5$ CFU·cm⁻² on the control surface and $\sim 2 \times 10^3$ CFU·cm⁻² on PFPA₀-IPPCL_{2.5}). Fewer bacterial colonies were observed on the surfaces of PFPA₅-IPPCL_{2.5}, PFPA₁₀-IPPCL₅, PFPA₁₀-IPPCL_{2.5} and PFPA₁₀ (all $< 1 \times 10^3$ CFU·cm⁻²). The antibacterial-adhesion efficiency of PFPA_x-IPPCL_y was calculated relative to the control, and the antiadhesion efficiencies on PFPA₅-IPPCL_{2.5}, PFPA₅-IPPCL_{2.5}, PFPA₁₀-IPPCL_{2.5} and PFPA₁₀ were above 99.99% against *V. alginolyticus*. These results indicate the exceptional antibacte-

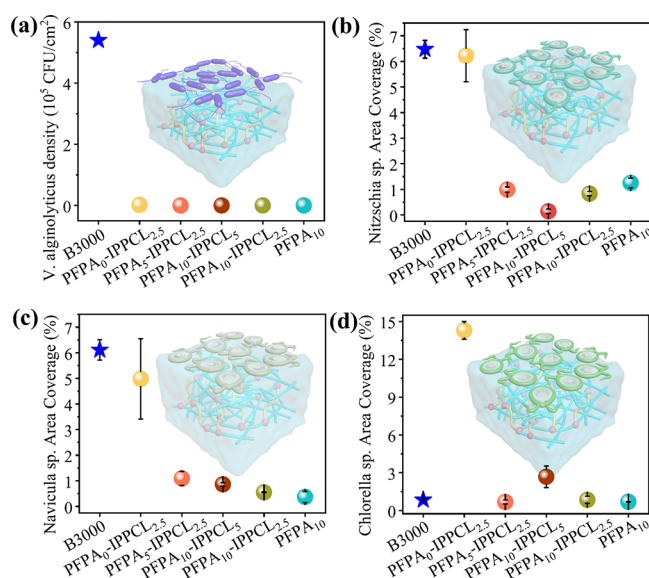


Figure 6. Bacterial densities of (a) *V. alginolyticus* on the LB medium after incubation with different coatings for 24 h. Area coverages of (b) *Nitzschia* sp., (c) *Navicula* sp., and (d) *Chlorella* sp. on coating surfaces.

rial-adhesion performance of the coatings. The number of colonies considerably decreased with the addition of lyophobic PFPE chain, indicating a prominent role of the low surface-free-energy components in bacterial-adhesion inhibition.

Algae constitute another representative class of marine fouling organisms that easily adhere to coating surfaces. To assess the degree of algal attachment to the coating surfaces, the area of red fluorescence from the chlorophyll in the cells was evaluated in this study. The distributions of algae cells on the coating surfaces are shown in Figure 6, and the area coverages are shown in Figure 6d–f. A large number of algal cells adhered to the control and PFPA₀-IPPCL_{2.5} surfaces and were aggregated in some areas. The area coverages of *Nitzschia* sp. and *Navicula* sp. were 6.47% and 6.12%, respectively, on the control surface and 6.22% and 4.98%, respectively, on the PFPA₀-IPPCL_{2.5} surface, indicating poor resistance to algal attachment. Lyophobic modification of the coating largely diminished the intensity of red fluorescence. This notable

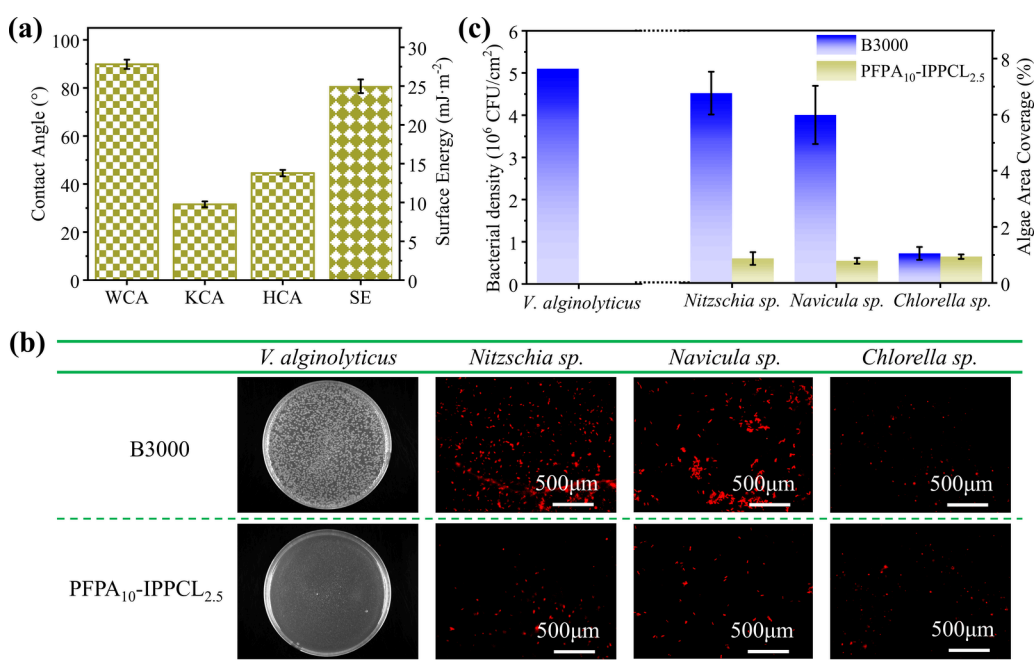


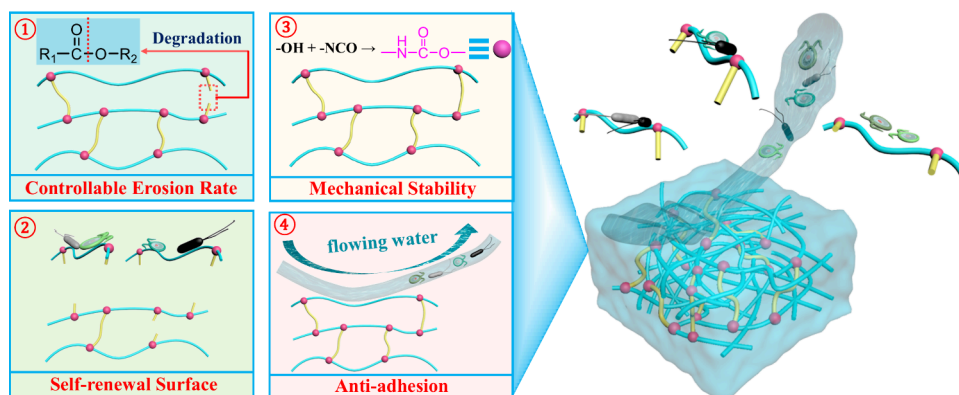
Figure 7. (a) Static contact angles of water, kerosene, and *n*-hexadecane on the PFPA₁₀-IPPCL_{2.5} surface after a 25-d erosion testing period. (b) Distribution of *V. alginolyticus* on the LB medium and distributions of *Nitzschia sp.*, *Navicula sp.*, and *Chlorella sp.* algal cells on the B3000 and PFPA₁₀-IPPCL_{2.5} surfaces after a 25-d erosion testing period. (c) Bacterial densities of *V. alginolyticus* on the LB medium after incubation for 24 h and area coverages of *Nitzschia sp.*, *Navicula sp.*, and *Chlorella sp.* on the B3000 and PFPA₁₀-IPPCL_{2.5} surfaces after a 25-d erosion testing period.

enhancement in antialgal adhesion capability demonstrates the deterrent potential of these coatings against algal growth. However, unlike the trends of antibacterial adhesion, the adhesion phenomena of algae on the coating surface were species-dependent. The area coverages of *Nitzschia sp.* on the PFPA₁₀-IPPCL₅, PFPA₁₀-IPPCL_{2.5} and PFPA₁₀ surfaces were 0.14%, 0.99% and 1.25%, respectively, exhibiting a monotonic decrease with decreasing SE. Meanwhile, the area coverages of *Navicula sp.* on the PFPA₁₀-IPPCL₅, PFPA₁₀-IPPCL_{2.5} and PFPA₁₀ surfaces were 0.86%, 0.55% and 0.37%, respectively, exhibiting a slight increase with decreasing SE. Therefore, the antidiatom adhesion performance of the coating can be compromised by an excessively low SE in some instances. Compared with B3000, PFPA₁₀-IPPCL_{2.5} reduced the adhesions of *Nitzschia sp.* and *Navicula sp.* by 84.6% and 91.0%, respectively. The area coverages of *Chlorella sp.* on the PFPA₀-IPPCL₅, PFPA₅-IPPCL₅, PFPA₁₀-IPPCL₅, PFPA₁₀-IPPCL_{2.5} and PFPA₁₀ surfaces were 14.3%, 0.68%, 2.68%, 0.84%, and 0.69%, respectively. Surprisingly, the antiadhesion performance of PFPA₁₀-IPPCL_{2.5} did not improve significantly compared with that of B3000, on which the area coverage was only 0.85%. Nevertheless, the low area coverage of *Chlorella sp.* indicates that PFPA₁₀-IPPCL_{2.5} can effectively prevent algae adhesion. Overall, PFPA₁₀-IPPCL_{2.5} exhibited exceptional antialgal adhesion resistance. Note that although PFPA₁₀ exhibits remarkable antibacterial and antialgal adhesion properties, its excessively high erosion rate renders it impractical for applications.

3.6. Stability of Antifouling Performance. Our coating system is continuously polished via the hydrolysis and cleavage of polycaprolactone polymer chains during applications. Therefore, conducting an in-depth analysis of the surface characteristics of the coating after the erosion test and evaluating its antifouling performance are important. Given the excellent antifouling performance demonstrated by

PFPA₁₀-IPPCL_{2.5} in previous studies, we focus on the evolution of the properties of this coating after 25-d erosion testing period. The static WCA of PFPA₁₀-IPPCL_{2.5} decreased to 89.8°, static HCA slightly decreased to 44.6° and SE increased correspondingly to 24.97 mJ·m⁻² after the erosion test (Figure 7a). This change can be attributed to hydrophilic carboxyl groups generated during the hydrolysis process, which decrease the static CA and increase SE. Figure S14a displays the full XPS spectrum of PFPA₁₀-IPPCL_{2.5} after 25-d erosion testing period. The spectrum shows the characteristic peak of Na 1s in addition to the characteristic peaks of C 1s, N 1s, O 1s, F 1s, Si 2s and Si 2p. Further insights into the surface properties of PFPA₁₀-IPPCL_{2.5} are obtained from the high-resolution C 1s, N 1s, F 1s and O 1s spectra (Figure S14 b–e, respectively) after 25-d erosion testing period. The quantitative analysis results for the atomic fractions in the probed surface regions are summarized in Table S2. The results show that erosion test does not affect the intensity of the F 1s peak, indicating that perfluoropolyether chains with low SE continue to accumulate on the surface after the test. The high-resolution O 1s XPS spectrum of the PFPA₁₀-IPPCL_{2.5} surface after 25-d erosion testing period can be deconvoluted into three peaks at 535.8 eV (–OH), 533.6 eV (C=O) and 532.0 eV (C–O). The peak at 535.8 eV indicates the break of ester groups and the formation of hydroxyl and carboxyl groups, which result in a decrease in SE. Concurrently, the C 1s spectrum of PFPA₁₀-IPPCL_{2.5} after the erosion test can still be deconvoluted into four peaks. However, the peak intensity corresponding to C–C significantly decreased, while the total peak intensity of the peaks corresponding to C=O and O=C–O increased, indicating the cleavage of ester groups and the formation of hydroxyl and carboxyl groups.

Figure 7b,c shows the antifouling performances of PFPA₁₀-IPPCL_{2.5} and B3000 after 25-d erosion testing period. The bacterial colony density was high on the B3000 surface after

Scheme 2. Schematic Showing the Distinctive Properties of the Degradable Low-Surface-Energy Coating^a

^a① The erosion rate of the coating can be controlled through the degradation of ester groups; ② self-renewing surfaces of the coating can eliminate fouling organisms adhering to the coating; ③ mechanical stability of the coating is achieved through the reaction of hydroxyl groups with isocyanate groups to form urethane linkages; ④ low-SE characteristic of the coating prevents the attachment of fouling organisms.

the erosion test (*V. alginolyticus* densities were $\sim 5.1 \times 10^6$ CFU·cm⁻²). Meanwhile, fewer bacterial colonies were observed on the surface of PFPA₁₀-IPPCL_{2.5} (*V. alginolyticus* densities were $< 1 \times 10^3$ CFU·cm⁻²). Obviously, PFPA₁₀-IPPCL_{2.5} still exhibited high antibacterial-adhesion efficiency after erosion test. Furthermore, the area coverages of *Nitzschia* sp., *Navicula* sp. and *Chlorella* sp. on B3000 were 6.77%, 5.99% and 1.05% and 0.87%, 0.79% and 0.94% on PFPA₁₀-IPPCL_{2.5} after the erosion test, respectively. Meanwhile, the algal area coverages on B3000 and PFPA₁₀-IPPCL_{2.5} did not vary significantly after the erosion testing, indicating the remarkable antialgal adhesion resistance of PFPA₁₀-IPPCL_{2.5}. Overall, PFPA₁₀-IPPCL_{2.5} retains its excellent antifouling performance after erosion testing, which is suitable for long-term applications.

Scheme 2 presents the distinctive properties of the antifouling coatings designed in this study. First, the isocyanate group of IPPCL reacts with the hydroxyl group of PFPA, generating a urethane group and thereby facilitating the construction of a cross-linked network structure. This network structure constrains the mobility of the molecular chain, thereby ensuring the mechanical stability of the coating (Scheme 2 ③). Notably, the enrichment of perfluoropolyether chains on the surface effectively reduces the SE of the coating, and the construction of cross-linked networks further reduces SE, which effectively inhibits the adhesion of marine fouling organisms. When such organisms adhere to the coating surface, they are weakly attached and are therefore easily dislodged by the scouring action of flowing water (Scheme 2 ④). Simultaneously, PCL chains break due to the hydrolysis of ester groups in seawater, allowing for the coating to be polished via scouring by seawater. Moreover, the dynamic erosion rate can be controlled by optimizing the cross-linking degree (Scheme 2 ①). More importantly, a minimal number of marine fouling organisms persist on the coating due to the complexity of the marine environment, deteriorating the antifouling properties. Thus, a degradable cross-linked network is a promising solution to this problem. The degradation of ester groups in IPPCL promotes the formation of a self-renewing surface, facilitating the removal of residual fouling organisms (Scheme 2 ②). This innovation sustains the antifouling performance and enhances the long-term durability of the coating.

4. CONCLUSIONS

In summary, this study demonstrates the successful preparation of multifunctional marine antifouling coatings with different liquid repellency and cross-linking degree. Due to the reaction between IPPCL and PFPA to form a cross-linked network, the tensile strength of the coating is greatly improved. Moreover, the flexibility of the coating is improved due to the low steric hindrance of methylene groups in IPPCL. Therefore, the coating exhibits satisfactory mechanical stability and low roughness during dynamic polishing. Interestingly, the adhesion of the coating increases with the increase in cross-linking degree, which may be related to hydrogen-bond-rich urethane bonds formed by the reaction between IPPCL and PFPA. It should be noted that the coating can be continuously polished under the action of flowing water, forming a new surface due to the degradability of polycaprolactone in seawater. Moreover, the erosion rate of the coating can be controlled by the cross-linking degree. Furthermore, the enrichment of perfluoropolyether chains on the coating surface can enhance its liquid repellency, and the formation of a cross-linked network can further improve liquid repellency. The antifouling test results show that the coating has excellent antibacterial and antialgal adhesion properties. Thus, the synthesis of IPPCL and perfluoropolyether resins can potentially lead to the development of self-polishing low-SE antifouling coatings for applications in various maritime industries.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.4c19192>.

Supporting figures and tables including the schematic diagram of the erosion test device, ¹H NMR spectrum of PCL after heavy water exchange, atomic percentage of elements, etc. (PDF)

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Notes

The authors declare no competing financial interest.

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